

TAUROVO, Russian Federation

WMO: 280970

Lat: 59.5855N

Lon: 73.2891E

Elev: 60

StdP: 100.60

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.2	(c) -36.0	(d) -41.8	(e) 0.1	(f) -38.8	(g) -38.7	(h) 0.1	(i) -35.6	(j) 4.1	(k) -9.6	(l) 3.5	(m) -10.7	(n) 0.4	(o) 150	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	28.9	19.1	27.2	18.7	25.3	17.9	20.6	26.3	19.6	25.1	18.7	23.6	1.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	23.3	17.7	12.8	21.7	16.8	12.1	20.9	59.6	26.4	56.5	25.5	53.5	23.7	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.3	(b) 3.7	(c) 3.3	(e) -43.9	(f) 31.2	(g) 3.2	(h) 0.9	(i) -46.2	(j) 31.8	(k) -48.1	(l) 32.3	(m) -49.9	(n) 32.8	(o) -52.2	(p) 33.5		
(4) 4.3	(5) 3.7	(6) 3.3	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-19.8	-15.7	-7.6	1.4	6.8	15.2	17.4	14.1	7.6	-0.4	-11.2	-16.0
	DBStd	14.22	8.86	8.81	7.08	4.72	5.11	4.82	3.70	3.91	4.22	5.19	8.70	9.17
	HDD10.0	4463	924	718	546	261	128	13	1	8	98	323	637	807
	HDD18.3	6973	1183	952	804	509	357	114	61	138	323	580	887	1065
	CDD10.0	594	0	0	0	1	31	170	231	137	24	1	0	0
	CDD18.3	64	0	0	0	0	1	22	33	9	0	0	0	0
	CDH23.3	860	0	0	0	0	44	308	390	114	4	0	0	0
	CDH26.7	202	0	0	0	0	7	80	98	17	0	0	0	0
Wind	WSAvg	1.6	1.4	1.5	1.8	1.9	2.0	1.6	1.6	1.4	1.5	1.7	1.6	1.4
Precipitation	PrecAvg	705	45	36	47	43	55	72	70	81	59	68	72	58
	PrecMax	917	71	62	110	67	119	118	142	138	97	124	111	113
	PrecMin	565	7	6	7	7	23	39	23	38	18	27	27	23
	PrecStd	79	18	13	23	14	19	20	28	24	20	24	21	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	3.4	7.0	19.2	27.9	31.0	32.2	28.6	23.4	16.1	4.1	0.4
		MCWB	-2.0	1.3	1.8	12.4	15.6	19.8	22.1	20.1	15.4	11.0	3.3	-0.2
	2%	DB	-3.6	0.7	4.6	15.0	23.9	29.1	28.9	26.3	19.6	11.2	2.1	-0.9
		MCWB	-4.1	-0.3	0.9	8.5	14.2	18.8	19.6	19.0	14.3	8.4	1.1	-1.5
	5%	DB	-5.4	-0.8	2.8	12.4	20.8	27.1	27.4	24.2	17.0	8.9	0.7	-2.7
		MCWB	-5.9	-1.7	0.2	7.1	12.5	18.6	19.0	17.9	12.8	7.1	0.1	-3.3
	10%	DB	-7.2	-3.8	1.5	9.9	17.0	24.8	25.4	22.2	14.8	6.9	-0.3	-4.3
		MCWB	-7.7	-4.7	-0.4	5.5	11.0	17.4	18.4	16.9	11.9	5.4	-0.8	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.9	1.3	2.6	12.4	16.6	21.3	23.0	21.0	16.6	11.1	3.4	0.1
		MCDB	-1.4	2.4	4.8	18.6	26.4	27.8	31.4	25.8	22.0	14.8	4.1	0.3
	2%	WB	-4.0	-0.1	1.5	9.6	14.6	20.1	21.1	19.8	14.6	8.8	1.4	-1.4
		MCDB	-3.5	0.5	3.6	13.8	22.0	27.0	26.6	24.7	18.1	10.4	1.8	-0.9
	5%	WB	-5.9	-1.6	0.6	7.6	13.2	19.2	19.9	18.7	13.5	7.3	0.2	-3.1
		MCDB	-5.4	-0.7	2.2	11.7	19.9	25.3	25.2	23.0	16.4	8.9	0.6	-2.7
	10%	WB	-7.6	-4.7	-0.3	5.6	11.4	18.0	19.1	17.6	11.9	5.7	-0.8	-4.7
		MCDB	-7.1	-3.8	1.4	9.3	16.9	23.7	23.9	21.1	14.6	6.9	-0.3	-4.3
Mean Daily Temperature Range	5% DB	MDBR	8.4	10.5	11.7	11.1	11.8	12.3	12.4	11.9	9.6	6.6	7.9	7.4
		MCDBR	7.7	7.2	10.9	15.4	18.6	16.0	15.2	15.1	13.5	9.8	6.4	5.9
		MCWBR	7.5	6.5	8.1	9.7	9.8	7.2	6.8	8.2	8.0	7.1	6.0	5.7
	5% WB	MCDBR	7.5	7.1	9.0	14.2	16.3	14.0	13.5	13.6	12.6	8.8	6.2	5.6
		MCWBR	7.3	6.6	7.0	9.3	9.3	7.1	6.8	7.6	7.9	6.8	6.0	5.4
Clear-Sky Solar Irradiance	taub		0.269	0.309	0.364	0.387	0.394	0.405	0.422	0.415	0.365	0.335	0.287	0.259
	taud		1.994	2.009	1.938	2.058	2.279	2.325	2.271	2.266	2.361	2.275	2.069	1.938
	Ebn at Noon		478	660	732	801	832	828	800	773	756	646	452	332
	Edh at Noon		51	87	129	137	119	115	119	111	86	68	47	36
All-Sky Solar Radiation	RadAvg		0.51	1.44	2.86	4.25	4.84	5.38	5.15	3.74	2.35	1.09	0.47	0.27
	RadStd		0.05	0.12	0.22	0.43	0.45	0.49	0.56	0.34	0.26	0.14	0.10	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-368	N/A	N/A

Nomenclature: See separate page